

**AETHERMOORE RESEARCH PACKET**

# Chemistry-Native CLI and Space Chemistry Systems

A governed command-line evidence room for chemistry, space chemistry, and external scientific engine bridges.

space

chemistry

cli

nasa-cea

pahdb

receipts

**Abstract.** SCBE should not claim to replace established chemistry engines. The stronger claim is a governed chemistry-native CLI lane that can route scientific tools, preserve receipts, and separate private proof from public evidence.

**Claim boundary.** Does not claim official NASA, RDKit, Open Babel, ASE, or DeepChem compatibility until bridges are implemented and verified.

**Status:** Research brief and product-scoping note.

**Research grade:** Research Brief

**Review model:** Initial source mapping and claim-boundary pass.

[PDF](#) | [Source note](#)

## Source Text

# Chemistry-Native CLI and Space Chemistry Systems Research

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## Executive Takeaway

SCBE should not claim to be the only AI CLI that can touch chemistry. Strong chemistry tools already exist. The defensible claim is narrower and stronger:

> SCBE has a governed chemistry-native CLI lane that combines symbolic chemistry, STISTA atomic-token flow, GeoSeed orbital invariants, private-proof hashes, and future bridges to established scientific chemistry engines.

The moat is not replacing RDKit, Open Babel, ASE, DeepChem, NASA CEA, or PAHdb. The moat is a governed agentic shell that can route them, audit them, preserve receipts, and keep private proof separate from public evidence.

## Existing External Chemistry Tools Worth Bridging

| Tool                | What It Is                                                                                                                                   | SCBE Bridge Target                                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| RDKit               | Open-source cheminformatics toolkit with Python/C++ core, descriptors, 2D/3D molecular operations, ML descriptors, and database integration. | <code>scbe chem rdkit</code><br>canonicalize, descriptors, substructure search, fingerprints, molecule validation. |
| Open Babel / obabel | Command-line chemistry toolbox for converting, filtering, and manipulating chemical data across many molecular formats.                      | <code>scbe chem convert</code> , format detection, file normalization, pipeline receipts.                          |
| ASE                 | Atomic Simulation Environment for setting up, manipulating, running, visualizing, and analyzing atomistic simulations.                       | <code>scbe chem atoms</code> , simulation setup wrappers, calculator receipts, structure export.                   |
| DeepChem            | Open-source Python library for deep learning in drug discovery, materials science, quantum chemistry, and biology.                           | <code>scbe chem ml</code> , dataset featurization, model cards, benchmark routing.                                 |

# NASA / Space Chemistry Systems Worth Bridging

| System                        | What It Does                                                                                                                                                                                                    | SCBE Bridge Target                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| NASA CEA / CEA2022            | Chemical Equilibrium with Applications solves equilibrium product concentrations and thermodynamic/transport properties for complex mixtures; used for rocket performance, shocks, detonations, and combustion. | scbe space-chem cea wrapper for input decks, output parsing, provenance, and comparison reports.           |
| CEARUN                        | Web/online interface that facilitates use of NASA CEA; CEA remains a gold-standard combustion/rocket thermochemistry code.                                                                                      | Local input-template generator and browser/API adapter where allowed.                                      |
| NASA Ames PAHdb               | Database and analysis tools for laboratory and computed PAH infrared spectra used to interpret astronomical observations, including JWST spectra.                                                               | scbe astrochem pahdb fetch/cache/fit pipeline with source receipts.                                        |
| SAM / Curiosity               | Mars rover instrument suite using gas chromatography, mass spectrometry, tunable laser spectroscopy, ovens, and wet chemistry for Martian samples and atmosphere.                                               | Not direct control; use as design pattern for sample -> instrument -> spectral evidence -> claim boundary. |
| CheMin / Curiosity            | Chemistry and Mineralogy X-ray diffraction instrument for identifying and quantifying minerals in Martian rocks and soils.                                                                                      | Benchmark-style data interpretation lane for mineral/spectral classification, not rover control.           |
| Spacecraft Materials Selector | NASA software catalog item for spacecraft material selection expert support.                                                                                                                                    | Future scbe materials advisory adapter with export-control and provenance boundaries.                      |

# Safe Claim Boundary

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Current SCBE evidence supports:

- symbolic chemistry and molecule-like scoring,
- STISTA/atomic-tokenizer flow,
- chemical-fusion reconstruction,
- ternary chemistry proxy tests,
- GeoSeed hyperbolic orbital invariants,
- private-proof hash inventory.

Current SCBE evidence does not yet support:

- validated wet-lab synthesis planning,
- validated hazardous chemistry instructions,
- official NASA CEA compatibility,
- official NASA, RDKit, Open Babel, ASE, or DeepChem certification,
- autonomous spacecraft or rover instrument operation.

## Recommended CLI Branches

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### 1. feat/chem-cli-core

- scbe chem atomize "text" - scbe chem fuse "tokens" - scbe chem bonds --coords ... - scbe chem orbitals --json - scbe chem prove

### 1. feat/chem-interop-bridges

- Detect optional tools: RDKit, Open Babel, ASE, DeepChem. - Expose wrappers only when installed. - Emit receipts with versions, command, input hash, output hash, and claim boundary.

### 1. feat/space-chem-bridges

- NASA CEA/CEARUN input deck templates. - PAHdb cache/search/fit scaffold. - SAM/CheMin-inspired evidence workflow for sample/spectrum/claim reports.

### 1. feat/chem-safety-gate

- Deny or quarantine hazardous synthesis, energetic material, toxicity, or illegal-drug requests. - Allow benign educational, file-conversion, descriptor, and public-data workflows. - Put human approval before any wet-lab actionable output.

## Product Framing

SCBE should present chemistry as a command-line evidence room:

```
Human intent -> STISTA atomization -> chemistry/space tool adapted
```

That is more valuable than claiming to be a replacement chemistry package. It lets a weak or strong model use real chemistry tools safely by following explicit steps, examples, expected outputs, and audit trails.

## Middle-Layer Design

We should do both: keep SCBE's symbolic chemistry/STISTA layer and bridge to real scientific chemistry systems through a governed middle layer.

```
public/private proof packets
      ^
      |
Human/AI request -> SCBE Chem Middle Layer -> External engines/data
      |
      v
STISTA atoms / GeoSeed orbitals / symbolic bonds
```

### Layer A: SCBE Native Chemistry

This is the internal symbolic layer:

- STISTA atomic tokenization,
- chemical fusion reconstruction,
- tongue molecule/bond scoring,

- ternary chemistry proxies,
- GeoSeed orbital invariants,
- private-proof hash inventory.

This layer is useful for agent routing, semantic decomposition, claim boundaries, and safety reasoning.

## Layer B: Chem Middle Layer

This is the translation and governance layer:

- canonical request schema: intent, material\_or\_molecule, representation, operation, risk\_class, allowed\_engines;
- adapter registry for RDKit, Open Babel, ASE, DeepChem, NASA CEA, PAHdb, SAM/CheMin-style data lanes;
- deterministic receipt schema: input hash, normalized input, engine, version, command, output hash, safety decision, claim boundary;
- bidirectional mapping: STISTA atoms can propose routes, external engines can return validated structures/properties that re-enter STISTA as evidence atoms;
- private/public split: public artifact gets hashes and result summaries; private artifact keeps patent/internal license material local.

## Layer C: External Chemistry and Space Systems

This is where validated scientific tools live:

- RDKit for cheminformatics and descriptors,
- Open Babel for conversion and file normalization,
- ASE for atomistic structures/simulation setup,
- DeepChem for ML chemistry workflows,
- NASA CEA/CEARUN for equilibrium/rocket thermochemistry,
- PAHdb for astrochemistry spectra,
- SAM/CheMin-style workflows for sample/spectrum/mineral evidence.

## Why This Is Stronger

The middle layer lets SCBE act like a governed chemistry operating shell:

- weak models can follow a tool contract without hallucinating chemistry,
- strong models get real engines and audit trails,
- hazardous or wet-lab actionable requests can be denied/quarantined,
- website claims can cite executable local receipts and external-tool provenance,
- patent/private proof remains hash-linked without exposing the private text.

## First Implementation Target

Build `scbe chem bridge` as the middle-layer endpoint:

```
scbe chem atomize "oxidizer fuel equilibrium"
scbe chem bridge --engine rdkit --operation descriptors --smiles
scbe chem bridge --engine obabel --from smi --to sdf --input molecule
scbe chem bridge --engine cea --template rocket --fuel CH4 --oxidizer
scbe chem prove --public
```

Initial adapters should be capability-detection only unless the tools are installed. The benchmark should pass by proving graceful detection, safe denial/quarantine, STISTA fallback, and receipt generation.

## Sources

- RDKit official documentation: <https://www.rdkit.org/>
- RDKit overview: [https://www.rdkit.org/new\\_docs/Overview.html](https://www.rdkit.org/new_docs/Overview.html)
- Open Babel documentation: <https://openbabel.org/>
- Open Babel command-line documentation:  
[https://openbabel.org/docs/Command-line\\_tools/babel.html](https://openbabel.org/docs/Command-line_tools/babel.html)
- ASE documentation: <https://docs.ase-lib.org/index.html>
- DeepChem GitHub/docs: <https://github.com/deepchem/deepchem>
- NASA CEA software catalog: <https://software.nasa.gov/software/LEW-17687-1>
- NASA CEA 3.0 beta documentation: <https://nasa.github.io/cea/>
- NASA CEARUN introduction: <https://cearun.grc.nasa.gov/intro.html>
- NASA Ames PAHdb page: <https://www.nasa.gov/?p=484873>
- Astrochemistry databases: <https://www.astrochem.org/databases.php>

- NASA Laboratory Astrophysics and Astrochemistry:  
<https://www.nasa.gov/general/laboratory-astrophysics-and-astrochemistry/>
- NASA SAM overview: <https://ssed.gsfc.nasa.gov/sam/samiam.html>
- NASA Curiosity science instruments: <https://science.nasa.gov/mission/msl-curiosity/science-instruments/>
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<https://software.nasa.gov/software/MFS-31328-1>

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